Sangeon Jeon

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Education

Georgia Institute of Technology | Atlanta, GA
Master of Science in Electrical and Computer Engineering, GPA 4.00 /4.0 *January 2025 – Present*
Bachelor of Science in Electrical and Computer Engineering, GPA 4.00 /4.0 *January 2023 – December 2024*

Korea Advanced Institute of Science and Technology | Daejeon, South Korea *February 2018 – February 2025*
Bachelor of Science in Electrical Engineering, GPA 3.95 /4.3

Work Experience

Dong Il Vision | Anyang, South Korea *June – August 2023*
Intern / Software Engineer

- Developed a C# WPF labeling tool to annotate images for machine learning.
- Contributed to building a computer vision system that differentiated products from competitors.

Research

Xiaoli Ma Lab @ Gatech | Atlanta, GA (Advisor: Prof. Xiaoli Ma) *Summer 2025 - Present*

- Conducting MATLAB simulations on Orthogonal Chirp Division Multiplexing (OCDM), a next-generation transmission scheme.

SIC-X Lab @ KAIST | Daejeon, South Korea (Advisor: Prof. Jinseok Choi)
Joint estimation of channel and mismatch for OFDM receiver with TIADCs *Fall 2024 - Present*

- Designed algorithms to estimate gain, timing, and offset mismatches in Time-Interleaved ADCs (TIADCs) under OFDM.
- Built a suboptimal maximum likelihood estimator with polynomial rooting, validated via MATLAB simulations.
- Achieved 84% reduction in computation time compared to existing methods while improving accuracy.
- Enhanced receiver calibration for high-speed OFDM systems, reducing mismatch-induced distortion in wideband communication channels.
- Currently preparing a research paper for publication.

Project

Korea Advanced Institute of Science and Technology | Daejeon, South Korea
KAIST Maker-a-thon (Topic: Item for the Elderly Living Alone) [link] *Summer 2024*

- Participated in three weeks of development, brainstorming and prototyping an item suitable for the topic using materials and equipment provided by KAIST makerspace. Led a team of four as the leader and won the top award with $2000.
- Proposed an RFID token system to manage the expiration dates of food in refrigerators. Innovated the pre-existing RFID refrigerator system by installing an RFID reader on a token container instead of the refrigerator.

Georgia Institute of Technology | Atlanta, GA
Senior Design Project: Plant Physiology [link] *Spring 2024*

- Created a system to record/play back Venus flytrap action potentials with Vivent (Switzerland).
- Implemented Seeeduino–MATLAB communication protocol and GUI app enabling reliable signal acquisition up to 100 Hz.

Awards

Roger Webb Award ECE Senior Scholar Awards *Spring 2025*
1st Prize at KAIST Maker-a-thon *Summer 2024*
Dean's List at KAIST & Gatech *Spring 2018, Fall 2022, Spring 2023, Fall 2023, Spring 2024*

Relevant Coursework

Korea Advanced Institute of Science and Technology | Daejeon, South Korea
Core Coursework: Signal and System, Probability and Introductory Random Processes, Linear Algebra, Communication Engineering
Georgia Institute of Technology | Atlanta, GA
Core Coursework: Fundamental of DSP, Random Signal Processing and Its Application, Application of DSP

Skills

Programming: MATLAB, Python (Pytorch, PyMC), C++
Hardware: Oscilloscopes, Function Generators, Multimeters, ATmega128A, Xiao Seeeduino
Languages: Korean (native), English (conversational), Japanese (intermediate), Chinese(beginner)